

Prompt

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Role & Task

- **Role:** A university professor specializing in Statistics.
- **Task:** Generate a comprehensive, plagiarism-free lecture slide for *Chapter 3* based on the official textbook. (Available at `prompt/bluman-step-by-step-statistics-8th-edition`).

Instructions

1. **Structural & Stylistic Alignment** Replicate the exact formatting, structural hierarchy, and stylistic layout of the provided slides (found in the `ch2` directory) to ensure visual and pedagogical consistency across all chapters.
2. **Comprehensive Mathematical Examples & R Integration**
 - **Exhaustive Extraction:** Extract and incorporate **every single example** from the Chapter of the textbook. Do not summarize or skip any examples.
 - **Dual-Solution Workflow:** For every example, provide a thorough, step-by-step solution featuring two distinct components:
 - **Mathematical Solution:** Write out the complete algebraic derivations, formulas, and manual calculation steps using standard LaTeX formatting (e.g., $\bar{x} = \sum_n^x$).
 - **R Implementation:** Provide the corresponding that solves the exact same problem (e.g., `xbar = sum(x)/length(x)`).
 - **Execution Outputs:** Ensure the R code is fully functional so that when rendered, both the code and its real execution results (e.g., vectors, plots, data frames) are explicitly displayed, mirroring the reference slides in `ch2/`.
3. **Paraphrasing & Academic Integrity** To prevent plagiarism, you must completely rewrite and paraphrase all textual descriptions, definitions, explanations, and commentary from the textbook. The narrative flow, tone, and wording must be entirely original, rigorous yet accessible, and tailored specifically for a student audience, while strictly preserving the technical and mathematical accuracy of the original content.
4. **Strategic Use of Callout Blocks** Utilize visually distinct **callout blocks** (e.g., alert/info/note boxes) to segment key information and enhance readability. Specifically, apply callout blocks in the following scenarios:
 - **Definitions & Formulas:** Whenever the textbook presents a standard definition box or a key mathematical formula.
 - **Example Prompts:** Enclose the initial question/prompt of every example within a callout block to separate it clearly from its solution and R code.
5. **Directory and Storage Architecture** Replicate the established file structure utilized in the previous modules. Dynamically generate and store the output within the newly designated directory (e.g., `ch3/`),

precisely mirroring the organization, sub-folders, and naming conventions of the repository.